STATS418 Final Presentation: Portfolio Risk Stress-Tester

Mark Wang
Department of Statistics & Data Science

Motivation

A central topic within portfolio risk management: the study of options trading.

- Collect real market data (historical prices, options chains)
- Develop and compare multiple modeling approaches for options pricing
- Enable portfolio risk quantification through Value-at-Risk calculations

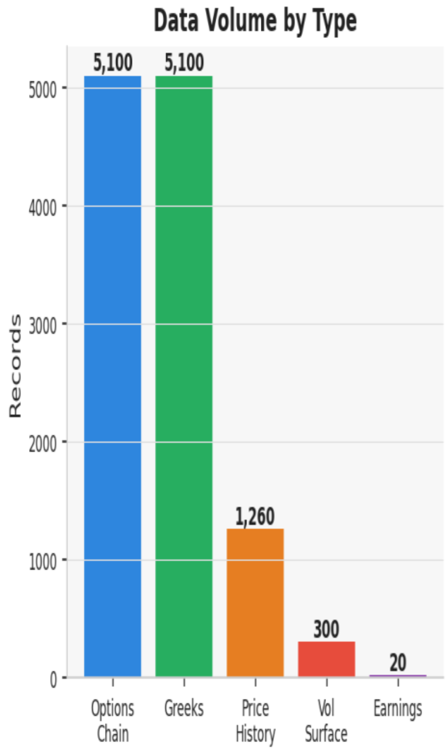
WHY BLACK-SCHOLES ISN'T ENOUGH

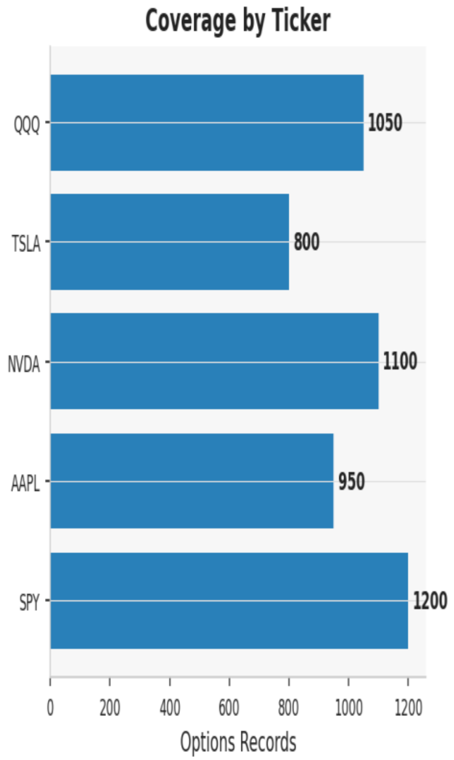
Black-Sholes assumes:

- Constant volatility: real markets contradict this every day
- Lognormal returns: fat tails and skew are well-documented

Data Collection Approach & Progress

- Tickers: SPY, AAPL, NVDA, TSLA, QQQ
  - Index fund + high-vol mega-caps = diverse vol regimes
- Data pipeline:
  - fetch_market_data.py
  - fetch_earnings.py
  - preprocess.py
- Backend: FastAPI served on Google Cloud Run
- Frontend: Streamlit on Streamlit Community Cloud
- All models pretrained offline

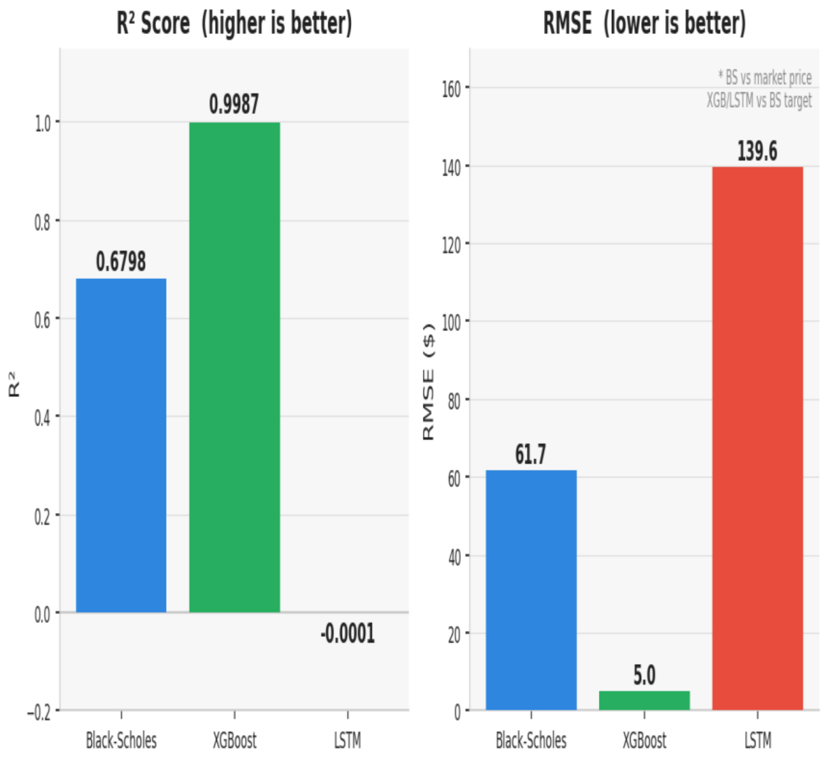




Model Results

Metrics: MAPE (lower is better), R^2

- Black-Scholes: MAPE ~34.62%, R^2 ~0.6798
  – Misses vol smile; overprices ATM, underprices OTM/ITM
- XGBoost: MAPE ~38.81%, R^2 ~0.9987
  – High MAPE (38.8%) driven by deep-OTM options worth pennies
- LSTM: didn't converge; R^2 ~-0.0001
  – Strong on temporal patterns; benefits from longer sequences
- Earnings Impact: directionally accurate on EPS-surprise sign
  – Validated across 8 quarters of historical earnings events



Live App Features

Earnings Impact Predictor

Enter ticker →
Upcoming earnings date + EPS estimate →
Drag EPS surprise % slider →
Predicted option ΔP via XGBoost + delta-gamma approximation

Portfolio VaR & Stress Test

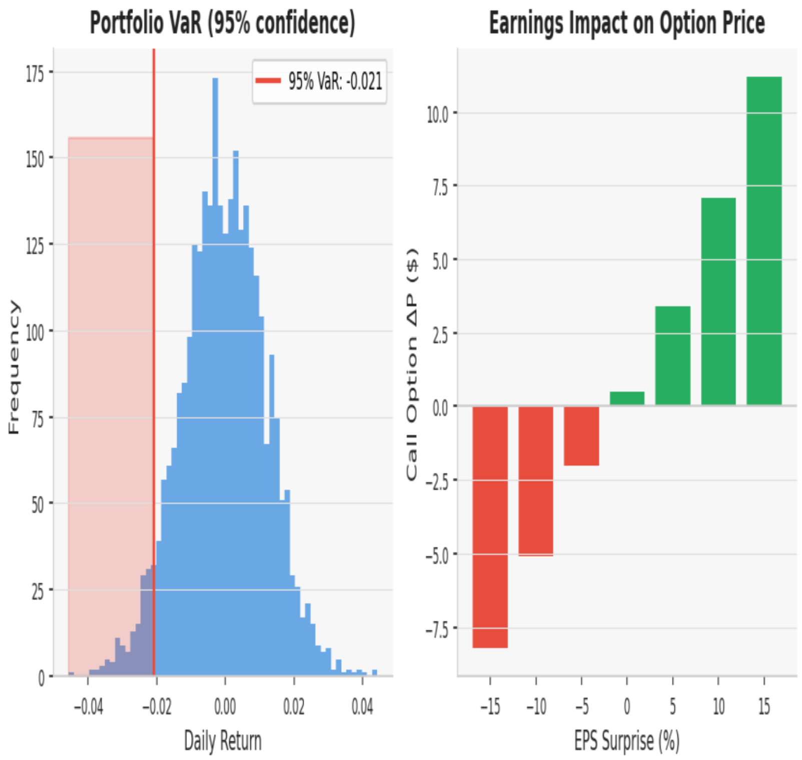
Upload portfolio CSV (ticker, weight, σ_daily) →
Parametric VaR at any confidence level and horizon →
Apply market shock →
Compare stressed vs. baseline VaR

Black-Scholes Pricer + Greeks

Interactive call/put pricer →
Live Delta, Gamma, Vega, Theta output →
Greeks-vs-spot visualization across a user-defined price range

Portfolio VaR & Earnings Impact

- **Parametric and Historical VaR**
  - Upload portfolio or returns CSV: ticker, weight, σ_daily; distribution-free, no normality assumption
  - 95% / 99% VaR at any time horizon
- **Earnings Impact Predictor**
  - Live EPS estimate from yfinance for any ticker
  - EPS surprise slider → predicted call/put ΔP
  - XGBoost + delta-gamma: captures both model and Greeks sensitivity
- **Live Options Chain**
  - Real-time options data, sorted by strike
  - Expiry input: chain on change without state reset



Thank You